CHAPTER 1

Overview of Mobile Computing

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Overview of Mobile computing

- **Mobile Computing** is the performance of computing tasks while the user is on the move, or visiting places other than their usual environment.
  - It is a technical field that covers the **design**, **development** and **evaluation of mobile applications** using appropriate solutions that meet user requirements.
- In mobile computing, users who are away from their ‘home’ intranet are still provided with access to resources via the devices they carry with them.
  - They can continue to access the Internet; they can continue to access resources in their home intranet; and there is **increasing provision for users to utilize resources** such as printers or even sales points that are conveniently nearby as they move around.
  - The latter is also known as **location-aware** or **context-aware computing**.

- **Ubiquitous computing** is the connecting of many small, cheap computational devices that are present in users' physical environments, including the home, office and even natural settings.
  - The term 'ubiquitous' is intended to suggest that small computing devices will eventually become so **pervasive** in everyday objects that they are scarcely noticed.
  - That is, their computational behavior will be transparently and intimately tied up with their physical function.

Cont'd

- Ubiquitous computing:
  - Activates the world,
  - Is invisible, everywhere computing that does not live on a personal device of any sort, but is in the woodwork everywhere,
  - Makes a computer so imbedded, so fitting, so natural, that we use it without even thinking about it.
- Also called: pervasive, deeply embedded, 4G mobile or sentient computing, and ambient intelligence.

Cont'd

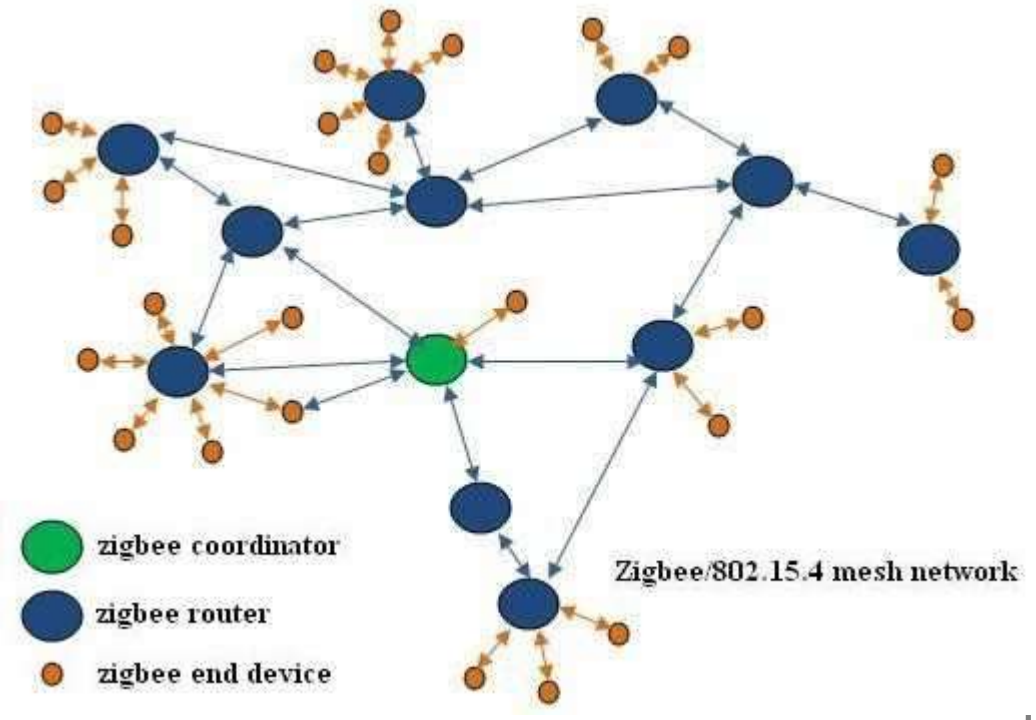
- Technological advances in device miniaturization and wireless networking
  - Led increasing integration of small and portable computing devices into distributed systems.
  - These devices include: portable and handheld communication devices connected using wireless networks.
    - Laptop computers.
    - Handheld devices, including mobile phones, smart phones, GPS-enabled devices, pagers, PDAs, video cameras and digital cameras.
    - Wearable devices, such as smart watches with functionality similar to a PDA.
    - Devices embedded in appliances such as washing machines, hi-fi systems, cars and refrigerators.
- The portability of many of these devices, together with their ability to connect conveniently to networks in different places, makes mobile computing possible and increasingly adopted and expanded throughout the world.

Cont'd

- **4 waves of computing**
  - **Mainframe computing (60's-70's)** – massive computers to execute big data processing applications – very few computers in the world.
  - **Desktop computing (80's-90's)** – one computer at every desk to help in business-related activities – computers connected in intranets to a massive global network (internet), all wired
  - **Mobile computing (90's-00's)** – a few devices for every person, small enough to carry around – devices connected to cellular networks or WLANs
  - **Ubiquitous computing (now)** – tens/hundreds of computing devices in every room/person, becoming "invisible" and part of the environment – WANs, LANs, PANs – networking in small spaces

Cont'd

- **PAN(Personal Area Network)**
  - A network which connects devices within small range typically 10 to 100 meters.
  - Device in one PAN network establish connection with another device in other PAN network when in the range.
  - Short range wireless technologies fall under PAN - **Bluetooth**, **Zigbee**, **Zwave**, Infrared etc.
- It is mainly designed and developed for low data rate monitoring and control applications.
- PAN wireless technologies have become very popular in IoT (Internet of things) networks.



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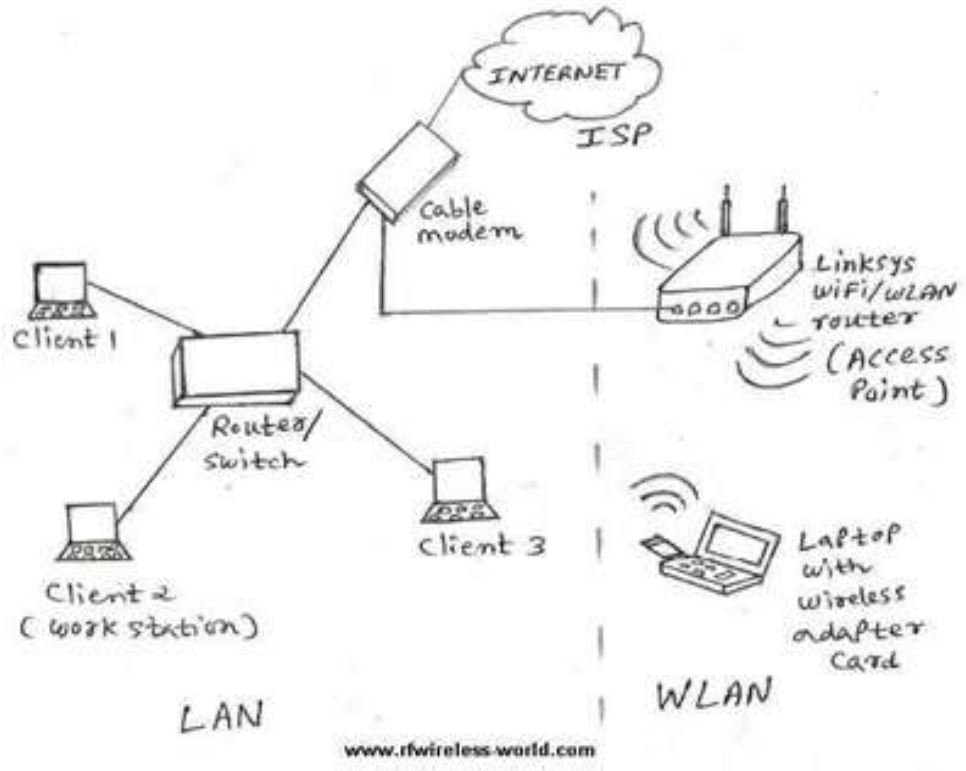
- **LAN and WLAN**
  - LAN is the short for Local Area Network.
    - A communication network which interconnects devices such as PCs, Laptops, Smartphones, tablets et. to exchange information.
  - LAN are of two types viz. wired and wireless.
    - Wired LAN - devices are connected using ethernet cables and all are wired onto ethernet Hub or Switch.
    - Wireless LAN - devices are connected using radio frequency waves which are electromagnetic waves.
  - Different bands of WLAN viz. 2.4 GHz, 5 GHz etc. WLAN have been as IEEE 802.11 family of protocols viz. 11a, 11b, 11g, 11n, 11ac, 11ad etc.



- **Mobile Computing Application**
  - The changes in technology change the order of the world in many ways such as **communication**, **business**, **provision of products and services**, etc.
    - Data shows that the penetration and adoption rate of mobile computing is high.
      - Use of mobile technologies has become part of daily life around the world as a way of sharing information.
        - For example, (in 2020) 96% of Americans own a cellphone of some kind whereas 81% of those are smartphones.
        - roughly half of the population own tablet computers and e-reader devices.
        - **In Ethiopia mobile penetration is 44.9 of every 100 inhabitant in the 2019/20 report.**
    - Therefore, mobile computing is the future of business because mobile technologies are used by most people around the world.
    - The implication is that government, public sectors and companies need to adapt application of mobile systems to provide their services, reach their customers, etc.

- Mobile computing technologies have wide application in every sectors:
  - **Health, financial, business, education, agriculture, tourism –**
  - **For payment, order, communication, booking, share information,** etc.
- In business, mobile technology influences the success of businesses in several ways, such as
  - **Simplification of administrative tasks** within a business.
  - To **conveniently and easily access business financial records** and documents whenever required (management/ stakeholders).
  - **Enhance communication and marketing**

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- **Mobile Computing - Features supported**
  - Support complicated security features such as
    - **Fingerprint recognition** systems and have security breach reporting features that assure data safety.
    - **Provide encryption** support to encode data stored in once company servers.
    - Provide a **biometric systems** – a security systems that are only accessed using biometric features, such as iris, fingerprints, and DNA, face, and voice recognition.
      - The biometric systems are more secure and they can work well with mobile devices like iPad and tablets that are commonly used in banks, hospitals, etc. to retrieve customer's records. E.g. Apple Pay
    - Companies are also moving away from the traditional hard disk data storage systems into **modern servers** (cloud computing services).
      - Advantages: offered on-demand, pooling of resources, scalability, increased performance, security, and efficiency of services. Eliminate the need to have network infrastructure.

Cont'd

- **Mobile Computing Limitation**
  - Mobile technology involves the flow of information within the enterprise and between the customers and the business.
    - Significant **risks associated with the flow of information**, especially in an unsecured mobile environment.
    - **Lack of comprehensive mobile solution** strategies exposes the enterprise it governance objectives.
    - For instance, the **cost of implementing the technology** might exceed the benefits or opportunities for the solutions adopted.
    - There are increased **chances of unaccountability** when using mobile solutions.
    - **Lack of policies and strategies** leads to miscommunication, inadequate management, and undetected inefficiencies.
    - **Lack of clear policies** affects business operations.

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- Mobile technologies require regular updates and support from the IT department.
  - Frequent upgrades and monitoring of the systems introduce the need to have an IT management team.
  - Lack resources or skilled employees to handle such changes.
- The nature of mobile devices means that they can easily be lost or stolen compared to their desktop counterparts.
  - Loss of mobile devices is also a security risk because malicious people can obtain critical data stored on these devices.
  - In case of organization, employees can also use their devices to store and transfer work-related information, which affects the confidentiality and integrity of data
  - Business continuity is affected during updates or when the technologies are disrupted by security risks.
  - Security risks in the mobile environment include cyber-attacks, hacking, malware attacks, disruption of network services, and human errors that expose valuable customer information
  - These activities might lead to increased cases of identity theft as hackers might sell information to third-party users or black markets.

Mobile Computing - Enabling Technologies

- Mobile computing hardware and software
  - Device
    - laptop, PDA, mobile phone, tablet, smart phone
  - Wireless(data) Communication
    - cellular telephony, data over cellular, wi-fi, Bluetooth, Zigbee, infra-red, 3G, 4G, 5G
    - higher bandwidth, lower power, commodity (readily available and secure)
  - System support – routing, billing, voice mail, data routing
- Small form factor devices
  - shrinking electronics, better displays, new input methods
- Personalization
  - Machine learning, Inference

Cont'd

- Automatic identification
  - RFID, numbering schemes, network information services
- Sensing and actuation
  - mechanical, chemical, electric, bio
- Context awareness
  - physical: properties of objects
  - Information: data, profile, provider
  - social: identity, situation, role
- Ambient displays
  - public screens, interaction
- Tangible interfaces

Cont'd

- Mobile Computing – **extremely varied**
  - Embedding for smart control
    - Embedded systems for cars, airplanes, etc.
  - Creating new computing devices
    - Hi-tech, silicon-based gadgetry, e.g. PDAs, cell phones, mp3 players, active displays
  - Connecting the existing physical world to a computational infrastructure
    - Ordinary objects and tasks re-evaluated and extended with computational/communication capabilities

Mobile Application Architecture

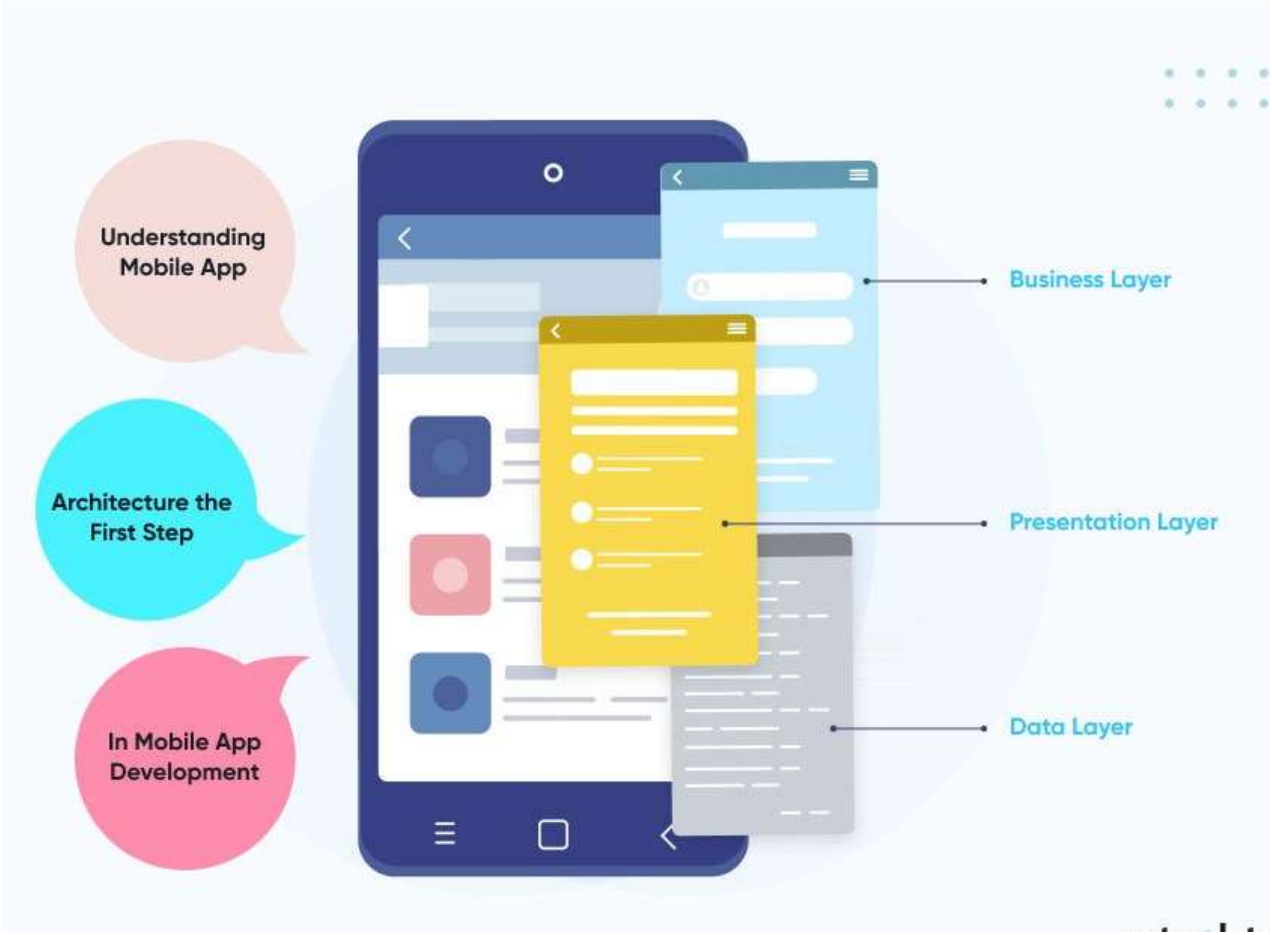
- Mobile app architecture refers to a **set of rules**, **techniques**, **processes**, and **patterns** to develop a mobile application.
  - These rules help developers create an app that meets both the business requirements as well as industry standards.

- **Mobile App Architecture vs Mobile Tech Stack**
  - The term mobile app architecture and mobile tech stack, although used interchangeable are different.
    - **Mobile technology stack** is the set of technologies and technical frameworks that make up the front and back-end of a mobile or web app (the what of the app), but is less concerned with the business / customer requirements (the why of the app) or the development process (the how of creating the app).
    - **The mobile app architecture** is made up of all the parts of the app – all the questions about why, what, how – including what data is collected, how the data moves, what the app looks like, for what platform, using what tech stack.

- **Value of mobile app architecture**
  - Many mobile apps developed today without any architecture or reference to standards.
  - Lack of architecture results in an app that is:
    - Longer and more costly to develop
    - Hard to maintain, particularly if staff change
    - Harder to build upon or scale
    - Difficult to test
    - More prone to errors
  - A good architecture is platform independent, but rather apply to native and cross-platform choices, resulting in a unified approach to development.
  - If we think of mobile app architecture as a skeleton for how we create a mobile app, we can then define layers (the bones, if you will) for how we build out the vital components of the app.

A Three layer – Mobile app architecture

- The most popular representation of mobile app architecture is represented by 3 layers: presentation, business logic, and data.



1. Presentation Layer of Mobile app architecture

- Presentation layer consists of all the processes and components to deliver the app to the user. When building the presentation layer, developers are concerned with what the user sees and feels when using the app.
- In other terms, the presentation layer is made up of the user interface (UI) and user experience (UX).
  - **User Interface (UI)** is concerned with design questions such as colors, fonts, placement, and overall design.
  - **User Experience (UX)** manages the way a customer interacts with the app through a detailed understanding of what a user wants and feels.
- When designing the presentation layer, developers need to determine the right platform and device type so that the presentation is consistent with the standards for each.

2. Business Layer of mobile app architecture

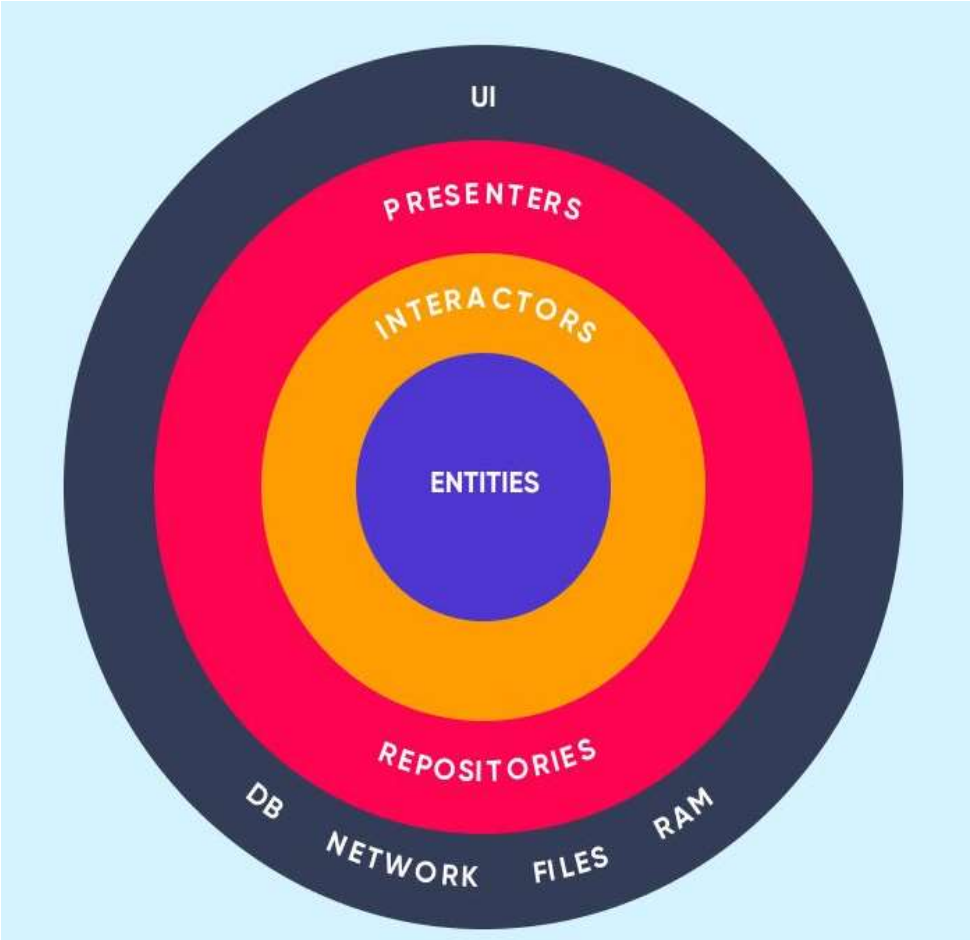
- The business layer is concerned with the logic and rules responsible for data exchange, operations, and workflow regulation.
- This layer is responsible for:
  - Security
  - Data caching
  - Logging
  - Data validation
  - Exception management
- The business layer can exist on the server or on the user device, depending on the operations of the app and the resources each operation takes.

3. Data Layer

- The data layer includes all the data utilities, service agents, and data access components to support data transactions.
- This layer can be thought of in two parts:
  - Persistence – data access with data sources via API
  - Network – network communication, routing, error reporting
- The design of the data layer must include thoughts about validation and maintenance of the data.

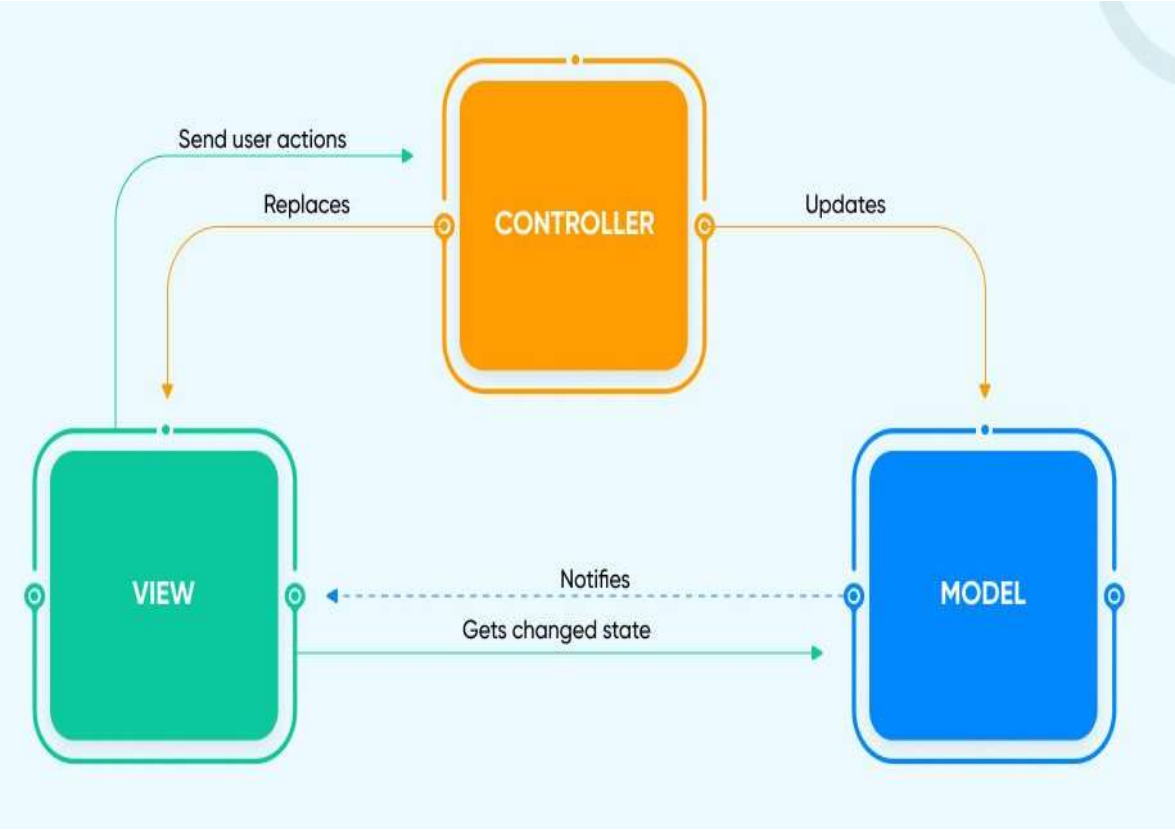
- **Android Mobile Application Architecture**
  - Apps developed specifically for Android are one kind of native app – an app developed for a specific mobile platform.
  - Android apps are developed to support Android languages (Kotlin and Java) for devices from a variety of manufacturers including Google, Samsung, Sony, and Nokia.
  - There is no a single architecture recommended for Android, but the most commonly accepted Android architecture for mobile apps is Clean Architecture.

- **Clean Architecture Manifest of Android**
  - In Clean, the architecture is built on the principles of layers and inversion of control.
    - Clean focuses on the same 3 layer structure modeled above, with the business layer sometimes referred to as the domain layer.
    - In Clean architecture, the domain / business layer must not depend on the other layers, but instead must leverage interfaces. While this can be difficult to understand, it does make it easy to add to and scale apps over time.

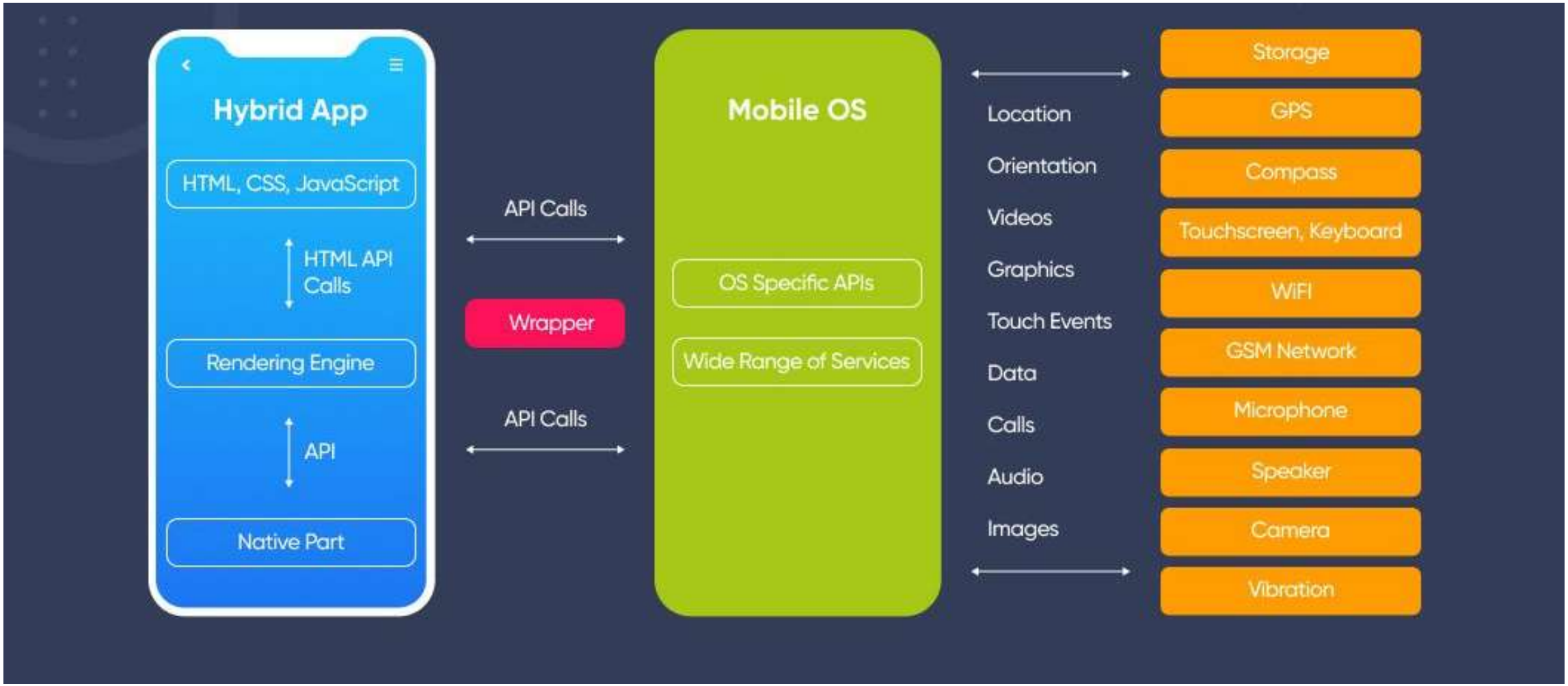


- **iOS Mobile Application Architecture**
  - Native iOS apps are developed using Objective-C and Swift languages, with Apple providing clear best practices over app architecture with the MVC model (Model-View-Controller).
  - While other options are available for iOS, the MVC model is made up of: model, view and controller (see the diagram in next slide).
  - The MVC model supports rapid and parallel development, with the ability to create multiple views – a great option for mobile development.

- **MVC model of ios app**
  - **Model** – The data layer (persistence, model objects, parsers, managers, networking code).
  - **View** – Similar to the presentation layer, a re-usable layer that represents the app to the user.
  - **Controller** – A mediator level that communicates with an abstraction via a protocol.



- **Hybrid Mobile Application Architecture**
  - Hybrid mobile apps leverage both native and web solutions. Hybrid apps use native apps as “shells” for the back-end, but platform-neutral JavaScript, HTML and CSS for the front-end.
  - Hybrid apps use plugins such as Apache Cordova or Ionic Capacitor to access native platform features.
  - Hybrid mobile apps are among the fastest apps to create across a variety of platforms and easy to update, but are not appropriate for complex, interactive, or feature-rich applications.


Hybrid App
HTML, CSS, JavaScript
HTML API Calls
Rendering Engine
API
Native Part
API Calls
Wrapper
API Calls
Mobile OS
OS Specific APIs
Wide Range of Services
Location
Orientation
Videos
Graphics
Touch Events
Data
Calls
Audio
Images
Storage
GPS
Compass
Touchscreen, Keyboard
WiFi
GSM Network
Microphone
Speaker
Camera
Vibration

Factors Considered While Developing Mobile App Architecture

1. Device Type

- When designing a mobile app, first choose the platform (iOS, iPadOS, Android, Windows, Cross-platform), then consider the different models of smartphone that are in use – and there are many.
- The following are important factors consider when designing a mobile app:
  - Screen size and DPI
  - Screen resolution
  - CPU (processor)
  - RAM (memory)
- The goal in this stage is trying to deliver the most consistent experience across various platforms and device sizes (tablet vs mobile, model varieties) so that every user – regardless of their device choice – has the best possible experience.

2. Development Frameworks

- Development frameworks are a consideration in designing the mobile app architecture as well as in establishing the tech stack.
- Frameworks provide libraries and basic templates and components for building mobile apps, both for the front and the back-end.
- For example,
  - The front-end frameworks for building mobile apps include bootstrap, foundation, react, angular, vue, and backbone.
  - The back-end (server-side), developmental frameworks depend on the chosen programming language and target platforms, including ruby on rails, flask, django, laravel, swift, xamarin, react native and flutter – among many others.

3. Bandwidth Scenarios

- User research is important to understand more about the target user.
- Around the world, users experience different bandwidth limitations, with some countries on 5G and others still experiencing spotty connections.
- For example, a highly interactive, graphic-heavy app will not be appropriate for apps that target rural users.

4. User Interface/ User Experience Design (UI/UX Design)

- When it comes to mobile app architecture, design plays an important role in first impressions (how it looks – a strong UI) as well as keeping users around (how it works – a strong UX).
  - A strong UX design is critical to online success, but mobile UX is complicated by changing user expectations and best practices of each operating system (OS) and device type.
- The mobile app architecture must balance UI against UX in the design phase.
  - Start by understanding the basics of mobile UX design and the latest mobile UX design trends to ensure the app is delivering value.

5. Navigation

- Navigation is the user's direct contact with the design, impacting both the front-end and the back-end.
  - A great mobile UX design helps users easily identify how to move around the page and explore further sections. Familiarity is key to navigation.
- Navigation best practices help ensure the mobile app is easy to use, reducing friction in the customer journey:
  - **Hamburger menu** – the three line menu in the navigation drawer or top bar is popular with mobile apps. Enhance navigation with a hover over menu, reducing clicks.
  - **Search** – A well-positioned search bar levels up usability, with standard position being the top right.
  - **Bars, rails, drawers, or tabs** – There are many options to navigate around an app, including fixed bars of buttons (top and/or bottom), rails (a vertical bar), drawers (hidden navigation), and tabs (screened content with fixed titles).

- **Familiar icons** – familiar icons such as home, search, photos, folder, etc. make navigation easier.
- **Intuitive labeling** – helping distinguish information with labels that spell out the intent of a button, option, or feature.
- **Site organization** – categorization can make site navigation easier – or more complicated.
- **Gestures** – supporting gesture-based navigation (swipes) can streamline navigation.
- **Scrolling** – how the app supports scroll vs fixed elements.
- And others.

Thumb Zones: Depicting the Ease and Difficulty of Thumb Touch on a Mobile Screen

6. **Push Notifications vs Real-time Updates**

- Mobile push notifications are messages that apps send to inform users they have a new in-app message, either from the app itself (such as product updates, offers, or reminders) or from another user.
  - Android users are automatically opted in for push notifications, but iOS users must opt-in, and typically do so at a lower rate (51% iOS vs 81% Android).
- There is a careful balance to be had between nudging users and annoying them when it comes to notification frequency and method.
  - Notification frequency can also have an impact on device battery life, which can in turn impact user retention.

- Mobile apps are leveraging real-time technologies to add greater interactivity to their apps.
  - For example, a delivery app can use real-time updates to provide updates about when a food order has been accepted, prepared, picked-up, or tracked on a live map.
  - Other examples include health and fitness apps or home technologies that benefit from real-time alerts (such as security systems).
- Like push notifications, real-time updates have their time and place in order to avoid overwhelming users with notifications.

Why You Should Use Mobile Push Notifications
We've prepared some interesting facts about push notifications that will help you promote your business app more effectively
Push notifications have an opening rate of 90%
Push notifications boost app engagement by 88%
09:20
70% feel that push notifications are useful
40% interact with push notifications within 1 hour of receiving them

Mobile Computing – Business Model

- Mobile computing is the future of business. It become the most profitable business in the digital mobile business.
- Many people enter the mobile business because they've become obsessed with the billion-dollar success of breakout apps like WhatsApp, Vine, Facebook and Instagram.
  - Mobile is a big deal, but to succeed one needs to build a sustainable mobile business.
  - Report disclose that about 2% of app developers claim approximately 54% of all app revenue.
  - All successful mobile business, definitely adopt some strategies or what we call business model, for their success.
- So, one has to plan or think of the strategy you can profit from the app, before you go to launch.

- The term **business model** is widely apply in the business sector to described the logic of a "business system" for creating value, that lies behind the actual processes.
  - It can be seen as a **detailed conceptualization of strategy** at an abstract level, which serves as a base for the implementation of a business processes.
  - A business model described as an **architecture composed of product, service and information flows**, including a description of the various business actors and their roles; a description of the potential benefits for the various business actors; and a description of the sources of revenues. [Similarly, in our case the business will be mobile app (start-up or lunch).
  - There are various types of business model mentioned in mobile app business. These includes: freemium, affiliate marketing to subscription and paid apps, etc.
  - Note that there is no hard and fast rule, to select specific model as every model has its advantages and limitations. Also, one can adopt more than one of these business models and integrate them.

1. Freemium

- This model offer **the app basic functionality for free**.
  - However, certain features are gated and cost money to be unlocked.
- In other words, people have access to an app's basic functionality, but there is a charge for premium or proprietary features.
  - The premise of this model is that you can **attract people to your app** and give them a rich preview of what your app can do (without giving them everything).
  - The goal is to accumulate and engage app users until they are willing to pay for additional in-app tools.
  - For the revenue, it offers the option for users to enhance their experience through purchases and subscriptions within the app, also called in-app purchases.
  - Note that only around 2% of the users will get converted in the best case, but the other 98% help you reach the potential audience. The reach makes the model very powerful and is widely popular for ***games and other mobile applications***
    - *Example Angry Birds. The Rovio team (the creator of Angry Birds) released a free version of their addictive app. However, the app kept certain features hidden until users upgraded (for a small fee) to the full version.*

2. Free, but with ads (in-app advertising)

- This model is used as a combination with a **free download of the application**, adding the option to purchase specific features, virtual goods, disable advertising, digital content and so forth.
- By offering related services return on investment can be made. This model is common in **gaming applications**.
  - It remove the cost-barrier to purchasing your app and allow free downloads.
  - The goal is to accumulate a large user base and gather information on the people interacting with your app. This data is then sorted and sold to app publishers who pay to place targeted ads in your app.
    - Facebook is the best example of this model. Its users don't directly pay Facebook anything to download or use their mobile platform, but Facebook leverages a vast amount of their data to sell highly targeted ads, a strategy which has proven to be effective for Facebook.

3. Advertisement

- This model offer the application completely free, however the return on investment is recovered by **introducing advertisement within the application**.
  - Digital advertisements have become widely prevalent and easy to integrate. Several platforms like Google Ads allow app developers to run ads in their app's interface and earn money out of it.
  - It is a great business model, and developers need not demand money from the users, while they can generate decent revenues.
  - Although, earnings per user is quite minimal unless too many ads are pushed. Doing that hampers the user experience and also drives them away. Hence, the success of this model relies on a wide user base.

4. Subscription Model

- This business model relies on users **buying the app subscription in exchange for its services**. With the rise of OTT(over-the-top)/steaming service apps like Netflix and Amazon Prime, this model has gained prominence quickly.
  - In simple words, the users will **pay a certain amount of money on a monthly** or **annual basis**, and the users can access the app properly, i.e, without being disturbed by ads.
  - This model is widely popular for ***video-streaming, music, and e-book*** related applications. You can also adopt a free trial period or keep some of the content free for everyone to lure more customers.

5. Affiliate Marketing

- Affiliate Marketing is a simple business model. Most e-commerce apps like Amazon and Flipkart **offer affiliate memberships** for their platforms.
  - If a sale is made through the affiliate member's link on the e-commerce website, the member receives a certain percentage as commission. It allows the app developers to earn decent money without hampering the user experience with ads or compel them to pay.
  - Nowadays, affiliate marketing has established its existence in several other types of businesses apart from e-commerce. This model can be easily applied by reviewers or apps that can influence decisions.

6. Paid Apps

- Both the well-known application stores in existence, the Google Play Store and App Store by Apple, allow app developers to list their apps on their platforms either for free or by **paying a certain amount of money**.
  - If you choose to make your app a paid one, you will earn money every time someone installs the app. Please note that both the App Store and Play Store charge 15% as commissions for all the sales made through their platforms for the first 1 million dollars each year.
  - If the revenues are more than a million dollars, Play Store charges 30% for everything beyond the million, while Apple shall levy 30% on the whole amount.
  - Although an attractive prospect, it limits your audience substantially. Developers should adopt this model only after proper market research based on extensive data.
  - There are also **Commissioned app**, **Sponsorships**, etc.

Questions?

Thank You